

Problem Statement and Goals

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Table 1: Revision History

Date	Developer(s)	Change
September 2026	SnowSight AI	Initial problem statement and project goals

1 Problem Statement

1.1 Problem

During winter season, municipalities are responsible for maintaining safe and usable transportation infrastructure. Winter maintenance activities, including snow clearing, plowing, and salting, are performed throughout large municipal road networks. Such services may be performed directly by municipalities or by independent contractors working on their behalf. Once maintenance has been completed, municipalities need to verify whether the final road condition satisfies the winter maintenance standard for that specific region.

The current records of maintenance activity do not clearly demonstrate the quality or completeness of the work performed. For instance, a record showing that a maintenance vehicle travelled along a road establishes that the vehicle was present, but does not prove that the road was left in an acceptable condition. Snow, ice, or debris conditions may still remain even after a maintenance pass. Therefore, municipalities need concrete evidence of the actual post-maintenance road condition instead of simply relying on records indicating that maintenance activity has occurred.

Verification is further complicated as acceptable road conditions are not always identical across municipalities. Different jurisdictions may define different levels of service, priorities, or acceptable post-maintenance requirements.

A road state that is considered acceptable under one municipality's standards may be considered unacceptable under another's. Evaluating completed maintenance work requires both an assessment of the resulting road condition and consideration of the standard applicable to the region in which the work was performed.

The sheer scale of municipal road networks makes consistent verification challenging. Manually examining large quantities of post-maintenance evidence requires significant time and effort and may result in conflicting assessments between different reviewers or road segments. Without a consistent assessment process, municipalities may have difficulty identifying inadequate maintenance, determining whether extra treatment is required, and evaluating whether a contractor has fulfilled the expected standard level of service.

Reliable verification is also essential for contractor accountability and payment decisions. Municipalities require sufficient evidence that contracted work has achieved the expected standard before determining whether the work qualifies for payment. Additionally, contractors have an interest in assessments being consistent, traceable, and based on the standards under which their work was performed.

The central problem is therefore the need for an objective, consistent, and scalable means of assessing the condition of municipal roads after winter maintenance, comparing that condition with the applicable municipal standard, and providing sufficient evidence to support maintenance-quality and contractor payment verification. The problem concerns assessing what condition the road was left in and whether that condition is acceptable; the particular technical method used to perform that assessment is outside the scope of the problem definition.

1.2 Inputs and Outputs

At a high level overview, the problem has three primary inputs.

The first input is evidence representing the observed condition of a road segment after winter maintenance services have occurred. This evidence represents the physical state in which the road was left and may contain observable conditions, like remaining snow, ice, slush, water, or other forms of debris.

The second input is the location information linked with the observed road segment. Location establishes which jurisdiction and road segment are being assessed and allows the appropriate maintenance expectations to be associated with the observation.

The third input is the related winter maintenance standard or level-of-service criterion for that location. This defines the conditions against which the observed road state is evaluated.

The primary output is a classification or assessment detailing the observed post-maintenance road conditions. The evaluation must provide a consistent and easily understandable representation of the road's condition.

A second output is a conclusion of whether the assessed road condition satisfies the applicable maintenance standard. This output separates work that

meets the relevant standard from work that fails to meet the standard or requires additional review.

A third output is a verification result that can assist in contractor-payment decisions. The result shows whether the available assessment supports the conclusion that the maintenance work satisfied the applicable level of service or whether the work should be referred for extra review. The final authorization and transfer of payment remain governed by municipal policies and contractual requirements and are not included in the core problem.

1.3 Stakeholders

1.3.1 Municipalities

Municipalities responsible for winter road maintenance are the primary stakeholders. Municipal staff need to verify whether roads have been maintained according to the standards set for their jurisdiction. They require clear and reliable information about post-maintenance conditions to assess service quality, identify work that may require additional attention, and support operational and contractual decisions.

1.3.2 Winter Maintenance Contractors

Winter maintenance contractors are in charge of performing services such as plowing, clearing, salting, and other treatments according to municipal and contractual expectations. Since assessments of completed work may impact performance evaluation and payment verification, contractors have an interest in the assessment process being consistent, transparent, traceable, and based on the applicable municipal standard.

1.3.3 Municipal VU Consulting Inc.

Municipal VU Consulting Inc. is the supervising organization and a project stakeholder. The organization contributes domain knowledge related to municipal winter maintenance operations, service standards, and maintenance quality verification. Its involvement helps ensure that the problem and resulting project goals reflect practical municipal needs.

1.3.4 Municipal Operations and Contract Administration Staff

Municipal staff responsible for winter operations, service monitoring, contract administration, and contractor management are expected to use the resulting assessment information. These users need to review road conditions, determine whether certain standards have been satisfied, investigate unsatisfactory results, and use the available evidence to support decisions.

1.3.5 Road Users

Drivers and other users of municipal road infrastructure are indirect stakeholders. They rely on municipalities and maintenance providers to keep transportation infrastructure reasonably safe and usable during winter conditions. Even though road users are not expected to interact directly with the system, improvements in the consistency of winter-maintenance verification can support the quality of the services on which they depend.

1.4 Environment

1.4.1 Physical Environment

The problem exists within municipal winter road networks during winter maintenance operations. Roads may be observed after plowing, salting, clearing, or other maintenance activities. Conditions can vary significantly between road segments and over time.

Post-maintenance road surfaces may contain certain amounts of snow, ice, slush, water, exposed pavement, accumulated winter debris, or combinations of these conditions. The surrounding environment may also contain vehicles, road markings, intersections, parked vehicles, snowbanks, and other infrastructure.

Environmental conditions may include active snowfall, precipitation, overcast weather, bright sunlight, darkness, reduced visibility, and changing illumination. These variations are part of the operating environment and make the appearance of otherwise similar road conditions non-uniform.

1.4.2 User Hardware Environment

Municipal staff are expected to review assessment results using regular desktop or laptop computers. Users are not required to have specialized processing hardware to review road-condition assessments, compliance results, or verification records.

1.4.3 User Software Environment

The user-facing information is expected to be accessible through commonly available modern web-browser software. Users should be able to review relevant road observations, assessment results, applicable standards, and verification information without interacting directly with the underlying processing components.

The development tools, programming languages, processing algorithms, and developer hardware used to implement the system are intentionally excluded from this section because they do not define the users' operating environment.

1.5 Scope

The core scope of the project is the verification of winter maintenance quality for municipal roadways after a maintenance or treatment activity has occurred. The

project focuses on assessing the resulting road state, determining whether that state satisfies the standard applicable to its location, and producing information that can support verification of contractor work.

The project does not replace municipal authorities in establishing winter maintenance standards. Instead, existing standards are treated as inputs to the assessment process. The project also does not make contractual policy or legal decisions on behalf of municipalities.

Contractor payment verification is within the project scope, but the actual transfer of funds is not part of the core system. The core system provides evidence and a verification result that can support a payment decision. An automated payment platform is treated as a stretch goal.

The core project focuses on roadways. Verification of sidewalks and bike lanes is treated as a stretch goal so that the primary project remains focused and achievable within the Capstone development period.

2 Goals

2.1 G1: Recognition of Post-Maintenance Road State

The system shall provide a consistent assessment of the observable condition of a road segment once winter maintenance has occurred.

The assessment shall highlight the differences between the road-state categories established for the project so that a post-maintenance observation can be assigned an interpretable road-state label. For this goal, success will be evaluated using a labelled set of representative road observations and an agreed-upon classification performance measure.

This goal provides municipalities with an objective description of the condition in which a road was left in instead of just confirming that a maintenance vehicle travelled along the road.

2.2 G2: Location-Specific Maintenance Assessment

The system shall verify whether an assessed road condition meets the winter maintenance standard applicable to the road's region.

Given a road-state assessment, its location, and the applicable municipal standard, the system shall generate a clear compliance result indicating whether the observed condition meets the relevant standard or requires additional review. Success for this goal will be measured by matching generated compliance results against expected results for a set of predefined road-state and municipal-standard scenarios.

This goal ensures that maintenance quality is evaluated according to the requirements of the responsible municipality instead of applying a single standard to each road.

2.3 G3: Maintenance Work Verification

The system shall output a traceable verification result for an assessed road segment that associates the road condition with the specific location, maintenance standard, and compliance result.

A municipal reviewer shall be able to determine from the verification result why the maintenance work was identified as meeting the applicable standard, not meeting the standard, or requiring further review. Success for this goal will be evaluated by concluding that verification records have the information necessary to trace the assessment to its associated observation, location, standard level, and compliance result.

This goal provides municipalities and contractors with reviewable evidence of how a maintenance-quality determination was attained.

2.4 G4: Contractor Payment Verification

The system shall use the maintenance compliance result to strengthen verification of whether completed contractor work satisfies the maintenance conditions associated with payment.

For an evaluated maintenance activity, the system shall provide a clear payment verification status indicating that the work is supported for payment, is not supported based on the applicable maintenance standard, or requires manual review. Success for this goal will be assessed using predefined maintenance and payment-verification scenarios with known predictable outcomes.

This goal is limited to aiding the payment-verification decision. Final payment authorization, contractual interpretation, and transfer of funds remain the responsibility of the municipality.

2.5 G5: Usable Review of Verification Results

The system shall present assessment and verification information in a form that municipal users can comprehend and review without requiring knowledge of the underlying processing methods involved.

A reviewer shall be able to identify the road segment, observed road state, applicable standard, compliance result, and payment-verification status for an assessment. Success for this goal will be assessed through representative review tasks where users must locate and interpret the information required to understand a verification result.

This goal ensures that the project's assessment capabilities produce useful information for the municipal stakeholders who ultimately need to review and act on the outcomes.

3 Stretch Goals

- **Contractor Payment Platform:** Extend payment verification into payment execution by permitting an approved contractor payment to be initi-

ated once the required municipal approval process has been finalized. This extends the core system beyond supporting payment-related decisions to facilitating the resulting transaction.

- **Dispute and Appeal Workflow:** Provide a structured process where contractors can dispute a maintenance assessment or payment-verification result. Municipal reviewers would be able to view the associated evidence, record a decision, and maintain a traceable history of the dispute.
- **Extended Verification for Sidewalk Lanes:** Extend maintenance-quality verification beyond only roadways to sidewalks and bicycle lanes. The extension would represent the different maintenance standards and observable conditions associated with these types of infrastructure.
- **Road Clearance Duration Estimation:** Estimate the duration for which an adequately treated road segment is expected to remain in an acceptable state by considering relevant environmental and treatment information. This could help municipalities in anticipating when additional maintenance may be required.

4 Custom Documents (CDs)

The Fall custom document will be the **Machine Learning (ML) Model Report**. This document will describe and evaluate the model used to support road-state recognition, including the data used for evaluation, the selected evaluation measures, experimental results, limitations, and relevant design decisions.

The Winter custom document will be the **Performance Report**. This document will evaluate the completed system against relevant performance objectives. The report will include quantitative measurements of system performance and discuss whether the implemented system provides adequate performance for its intended municipal use.

These custom documents may be refined with the approval of the course instructor as the project progresses.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?